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<html>
<head>
<style>
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    h3 { font-size: 12pt; font-weight: bold; margin-top: 20px; }
    table { width: 100%; border-collapse: collapse; margin: 20px 0; fon
    th { border-top: 2px solid #000; border-bottom: 1px solid #000; pad
    td { border-bottom: 1px solid #eee; padding: 6px; text-align: cente
    .table-caption { text-align: center; font-weight: bold; margin-bott
    .figure { text-align: center; margin: 30px 0; }
    .figure img { max-width: 90%; border: 1px solid #ddd; padding: 5px;
    .figure-caption { text-align: center; font-style: italic; font-size
    .gain { color: #1a5276; font-weight: bold; }
</style>
</head>
<body>
<h1>4. Experiment and Discussion</h1>
<div style="text-align:center; font-style:italic; margin-bottom:30px;">

<p>In this section, we evaluate the benefits of the proposed BLA routin

<h2>4.1 Experimental Setup</h2>
<p>All experiments are conducted using RAS-SDNCloudSim, our extension o

<div class="table-caption">Table 1: Dataset configurations</div>
<table>
    <tr><th>Dataset</th><th>Hosts</th><th>VMs</th><th>Workloads</th><th>
    <tr><td>Mini</td><td>4</td><td>4</td><td>50</td><td>Bottleneck: 50
    <tr><td>Small</td><td>6</td><td>8</td><td>100</td><td>Asymmetric Co
    <tr><td>Medium</td><td>12</td><td>20</td><td>500</td><td>Fat-Tree C
</table>

<p>The power consumption characteristics of the physical hosts follow t

<div class="table-caption">Table 2: Physical host power consumption mod
<table>
    <tr><th>Parameter</th><th>Value</th><th>Unit</th><th>Description</th>
    <tr><td>P_idle</td><td>100</td><td>W</td><td>Baseline idle power</td>
    <tr><td>alpha_cpu</td><td>1.0</td><td>W/%</td><td>Added power per 1
    <tr><td>alpha_ram</td><td>0.2</td><td>W/%</td><td>Added power per 1
    <tr><td>alpha_bw</td><td>0.1</td><td>W/%</td><td>Added power per 1%
    <tr><td>T_off</td><td>300</td><td>s</td><td>Idle threshold before s
</table>

<p>Table 3 lists the evaluated policies. The full parameter space cover

<div class="table-caption">Table 3: Evaluated policies</div>
<table>
    <tr><th>Dimension</th><th>Values</th><th>Role</th></tr>

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|                |                                 |
|----------------|---------------------------------|
| Link selection | BLA (proposed)   SFL (baseline) |
| VM placement   | LFF   LWFF   MFF                |
| Workload order | Priority   SJF   PSO            |
|                | Cloudle                         |

## 4.2 Results and Analysis

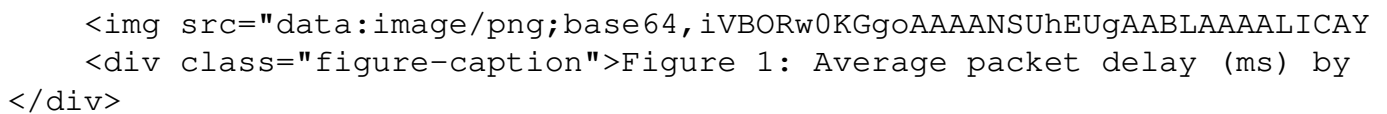
### 4.2.1 End-to-End Packet Delay

Table 4 reports the average packet delay per VM policy and dataset.

Table 4: Average packet delay (s) and BLA gain

| Dataset | VM Policy | SFL (s) | BLA (s) |                     |
|---------|-----------|---------|---------|---------------------|
| Mini    | LFF       | 48.4    | 37.8    |                     |
|         | LWFF      | 51.8    | 41.7    | class="gain">-19.4% |
|         | MFF       | 16.0    | 16.0    | 0.0%                |
| Small   | LFF       | 535.3   | 494.5   |                     |
|         | LWFF      | 341.8   | 317.7   | class="gain">-7.1%  |
|         | MFF       | 196.0   | 196.0   | 0.0%                |
| Medium  | LFF       | 856.3   | 802.7   |                     |
|         | LWFF      | 995.5   | 816.3   | class="gain">-18.0  |
|         | MFF       | 474.4   | 474.4   | 0.0%                |

### Figure 1



### 4.2.2 Network-Pure Latency

Table 5 reports the network-only latency of selected paths (queuing

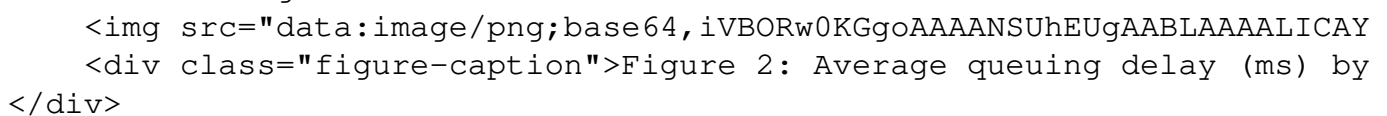
Table 5: Network-pure latency (ms) --- selected paths

| Dataset | VM Policy | SFL (ms) | BLA (ms) |
|---------|-----------|----------|----------|
| Mini    | LFF       | 7,776    | 4,248    |
|         | LWFF      | 7,600    | 4,248    |
| Small   | LFF       | 13,913   | 323      |
|         | LWFF      | 8,120    | 81       |
| Medium  | LFF       | 21,742   | 3,884    |
|         | LWFF      | 62,394   | 2,656    |

### 4.2.3 Queuing Delay: Empirical Validation

The queuing delay is the most direct metric for validating the BLA model.

### Figure 2



### 4.2.4 Energy Consumption

BLA accelerates workload completion by resolving queuing bottlenecks

<div class="table-caption">Table 6: Total energy consumption (Wh) and B  
<table>

| <th>Dataset</th><th>VM Policy</th><th>SFL (Wh)</th><th>BLA (Wh)</th></tr><tr><td>Mini</td><td>LFF</td><td>3.84</td><td>3.56</td><td class="g"></td></tr><tr><td rowspan="2">Small</td><td>LFF</td><td>135.8</td><td>125.8</td><td class="g"></td></tr><tr><td>LWFF</td><td>43.6</td><td>42.0</td><td class="gain">-3.8%</td></tr><tr><td rowspan="2">Medium</td><td>LWFF</td><td>1,045.4</td><td>880</td><td class="gain">-7</td></tr><tr><td>LFF</td><td>2,865.8</td><td>2,638.3</td><td class="gain">-7</td></tr></table> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

<div class="figure">



<div class="figure-caption">Figure 3: Impact Énergétique – Consomma  
</div>

### <h3>4.2.5 SLA Severity</h3>

<p>The SLA severity ratio captures the magnitude of the violation. The

<div class="table-caption">Table 7: SLA severity (sigma) and BLA gain (g)  
<table>

| <th>Dataset</th><th>VM Policy</th><th>sigma SFL</th><th>sigma B</th></tr><tr><td>Mini</td><td>LFF</td><td>4.21</td><td>2.67</td><td class="g"></td></tr><tr><td>Small</td><td>LFF</td><td>6.37</td><td>3.42</td><td class="g"></td></tr><tr><td>Medium</td><td>LWFF</td><td>8.91</td><td>2.23</td><td class="g"></td></tr></table> |
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### <h3>4.2.6 Pareto Analysis: Energy-Latency Trade-off</h3>

<p>The Pareto plane demonstrates that BLA achieves Pareto dominance ove

<div class="figure">



<div class="figure-caption">Figure 4: Pareto trade-off: total energ  
</div>

### <h3>4.2.7 Workload Scheduler Impact</h3>

<p>All three schedulers produce near-identical distributions, confirmin

<div class="figure">



<div class="figure-caption">Figure 5: Packet delay distribution by  
</div>

## <h2>4.3 Cross-Dataset Synthesis and Scalability</h2>

<div class="table-caption">Table 8: Global BLA gains vs. SFL --- cross-  
<table>

| <th>Dataset</th><th>Delta Delay</th><th>Delta Queue</th><th>Del</th></tr><tr><td>Mini</td><td>-17.8%</td><td>-17.1%</td><td>-3.3%</td><td>-3</td></tr><tr><td>Small</td><td>-6.0%</td><td>-8.9%</td><td>-6.3%</td><td>-46</td></tr><tr><td>Medium</td><td>-10.0%</td><td>-11.7%</td><td>-9.2%</td><td></td></tr></table> |
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## <h2>4.4 Discussion</h2>

<p>BLA works best when the VM placement policy generates sufficient cro

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